

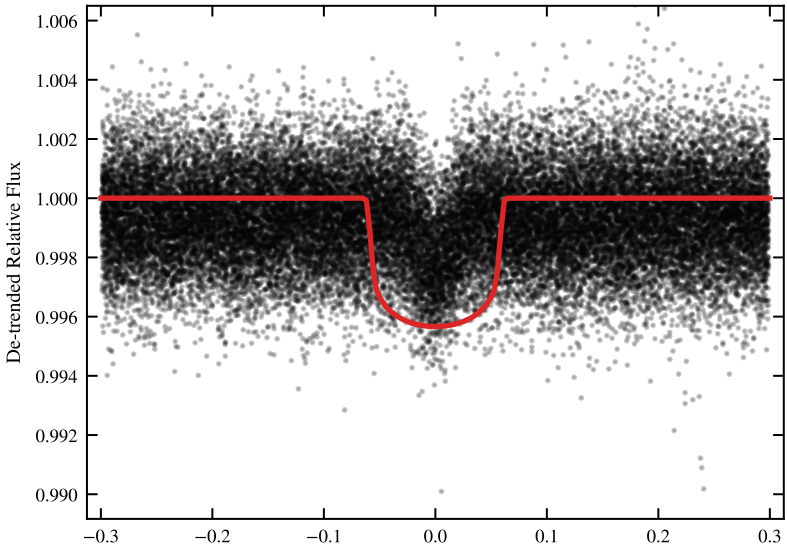
TESS PLANET TRANSIT CHARACTERIZATION REPORT

Target: TIC 136274063 | RA: 230.49883° | Dec: -53.81960° | Sectors: 102, 103

Derived Planetary Parameters

Parameter	Derived Value (MCMC)
Period (d)	$0.59713 \pm 3.48\text{e-}10$
Rp/R★	$0.060909 \pm 1.84\text{e-}07$
Rp (R⊕)	15.2 ± 0.568
b	$0.1 \pm 3.78\text{e-}07$
T14 (h)	$3 \pm 3.14\text{e-}06$
a (AU)	0.018393 ± 0.000687
Teq (K)	4144 ± 87.2

Primary Phased Transit



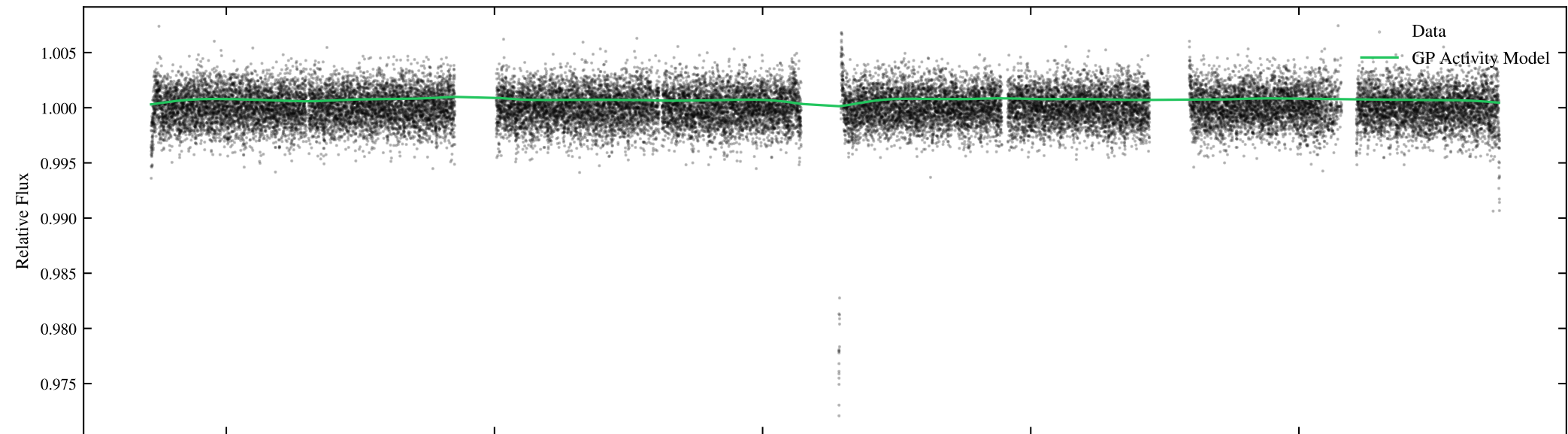
Literature Comparison & MCMC Diagnostics

Parameter / Metric	Literature Value	Derived / Diagnostic
P1 Period (d)	0.597131	0.597131 ± 0.00000
P1 Epoch (BTJD)	N/A	2459360.1605 ± 0.0000
Max R-hat (R)	N/A	3.7828
Min ESS	N/A	4
MCMC Divergences	N/A	0

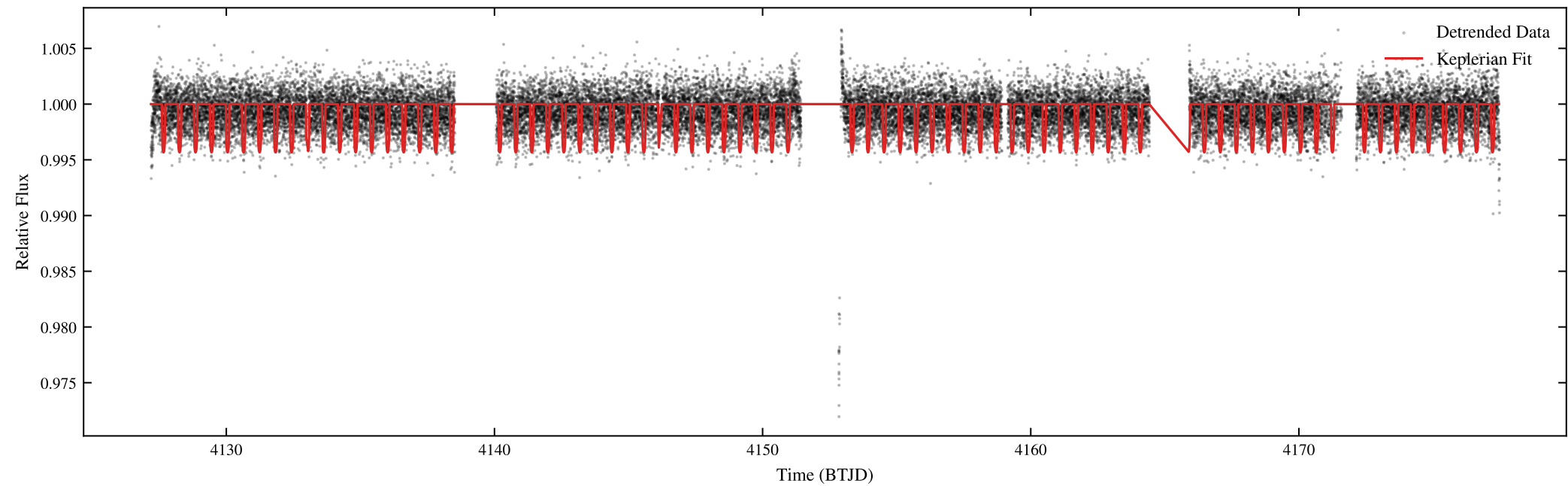
Host Star Properties

Stellar Parameter	Value	Source
R★ (R☉)	2.285 ± 0.086	VizieR TIC8
M★ (M☉)	1.81 ± 0.226	VizieR TIC8
Teff (K)	7710 ± 161	VizieR TIC8
log g (dex)	3.978 ± 0.0522	VizieR TIC8
[Fe/H]	N/A	Catalog
ρ★ (g/cm³)	0.2139 ± 0.036	Derived

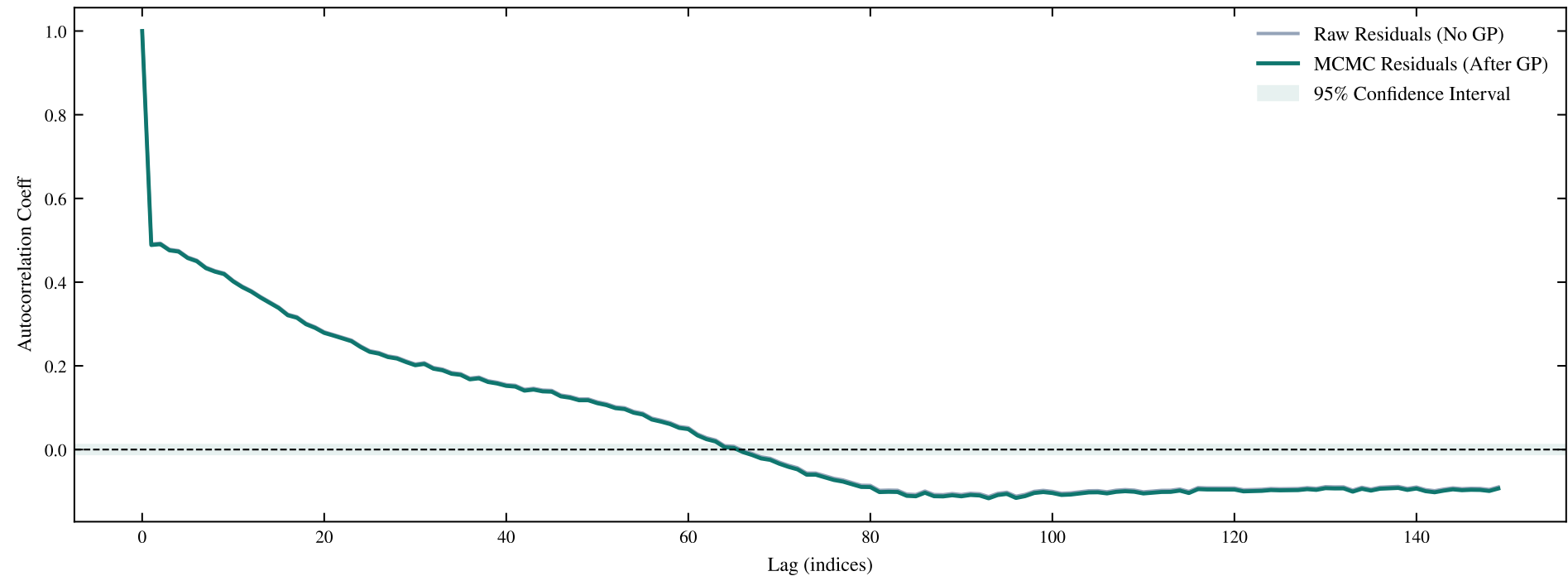
TIC 136274063 | Sectors: 102, 103 | Bayesian GP Detrending (Time Domain)



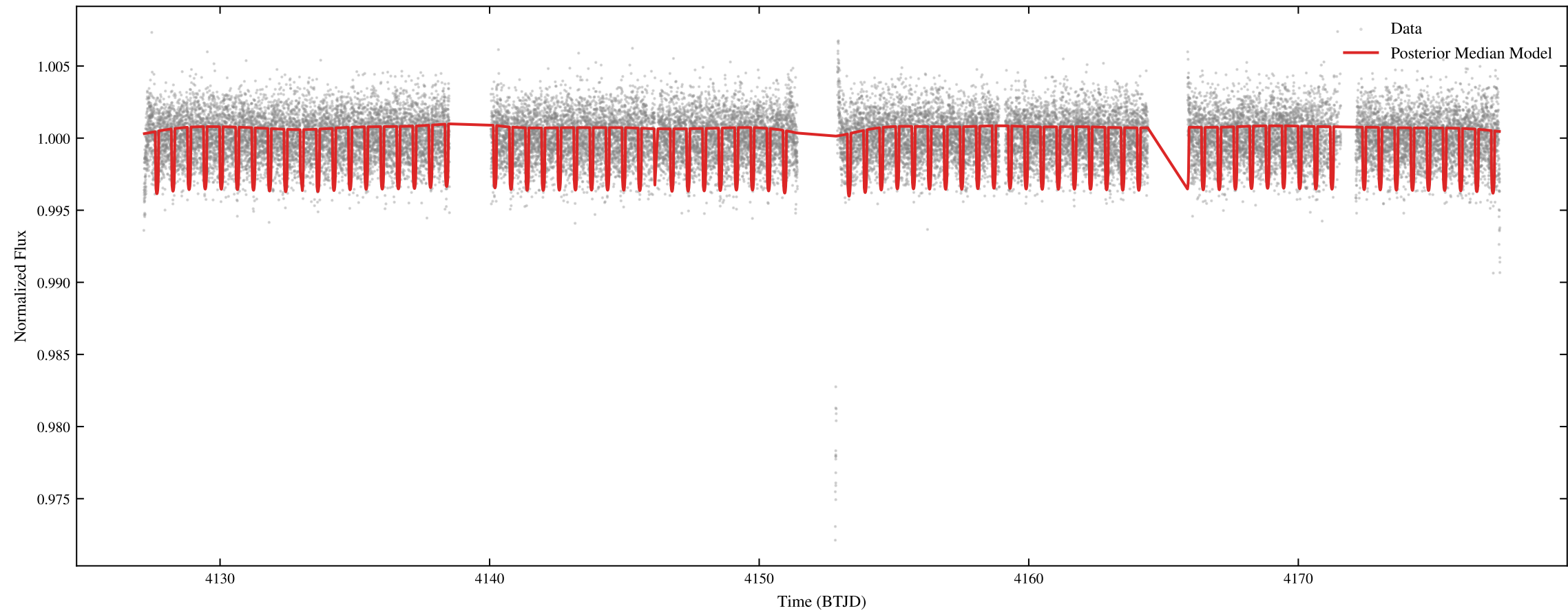
Detrended Light Curve & Best-Fit Transit Model

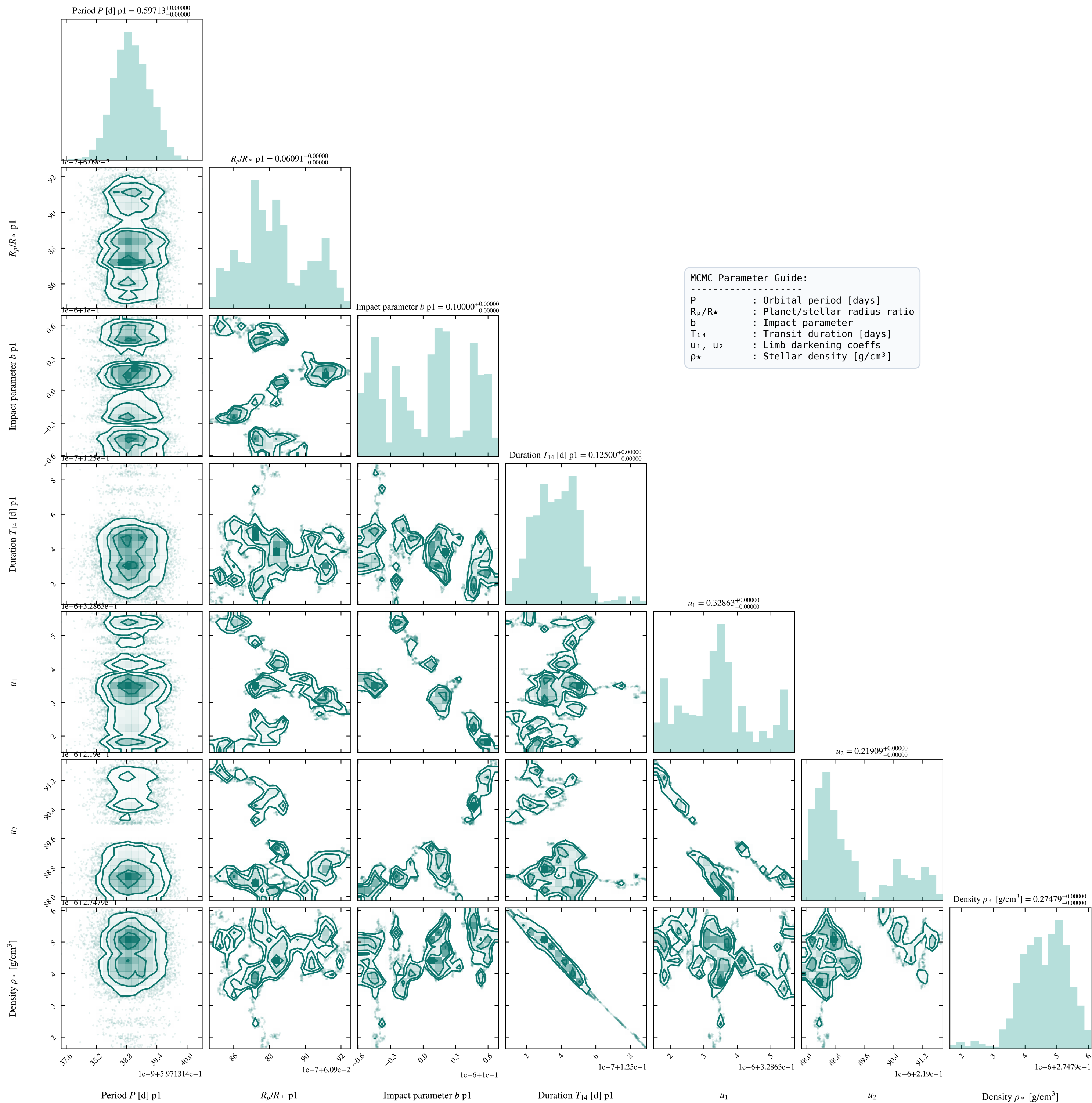


TIC 136274063 | Sectors: 102, 103 | GP Residuals Autocorrelation Function (ACF)

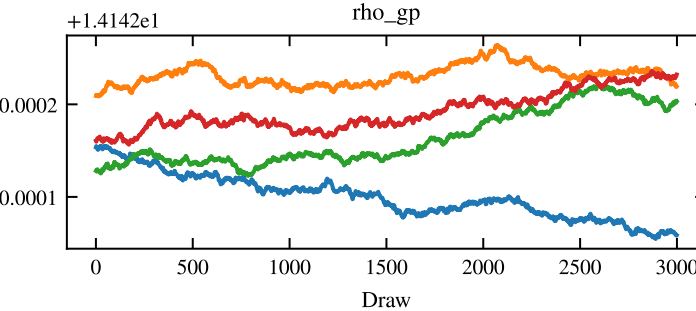
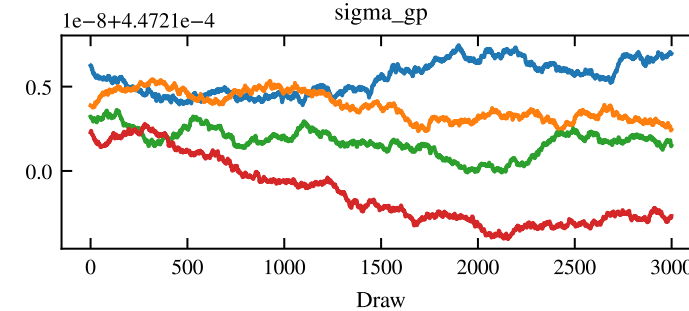
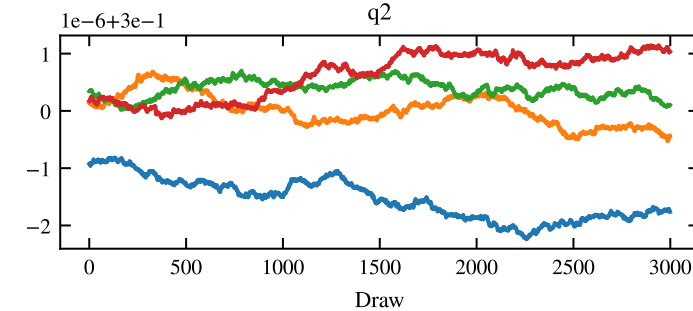
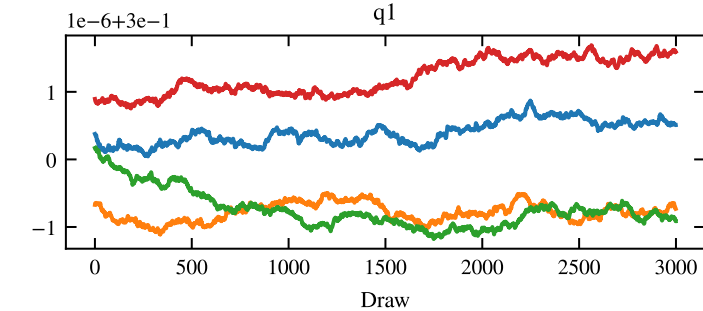
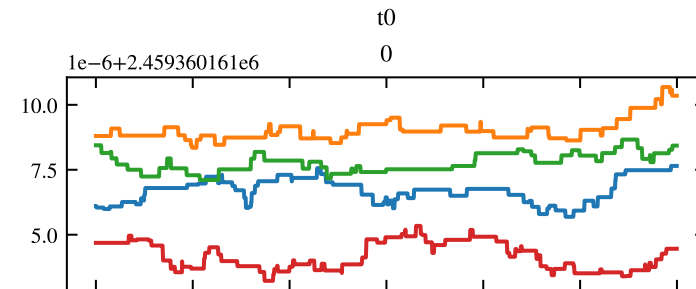
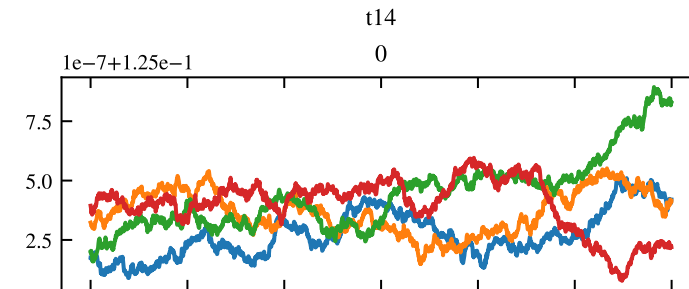
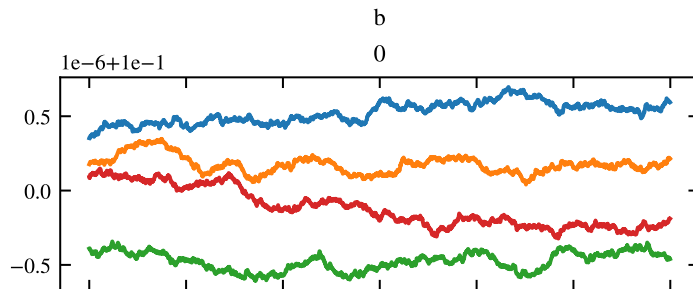
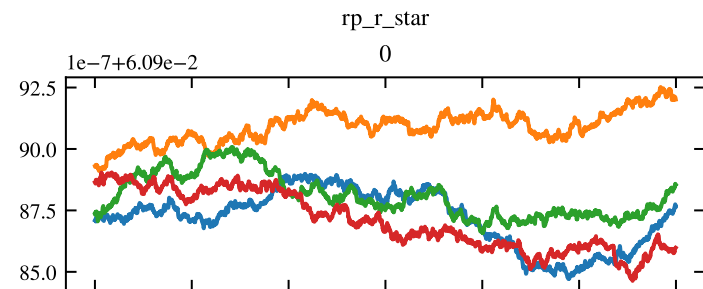


TIC 136274063 | Sectors: 102, 103 | MCMC Posterior Predictive Check

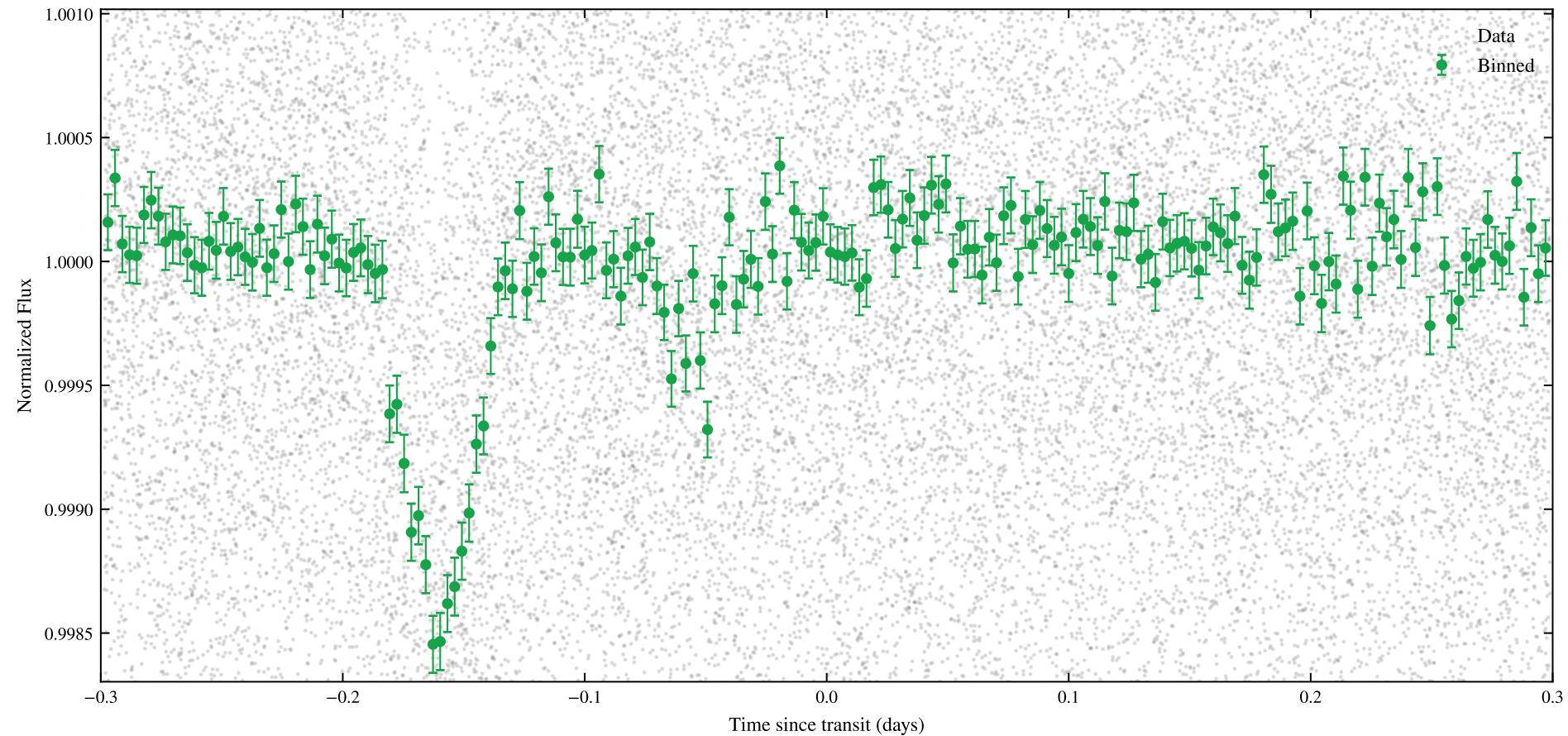




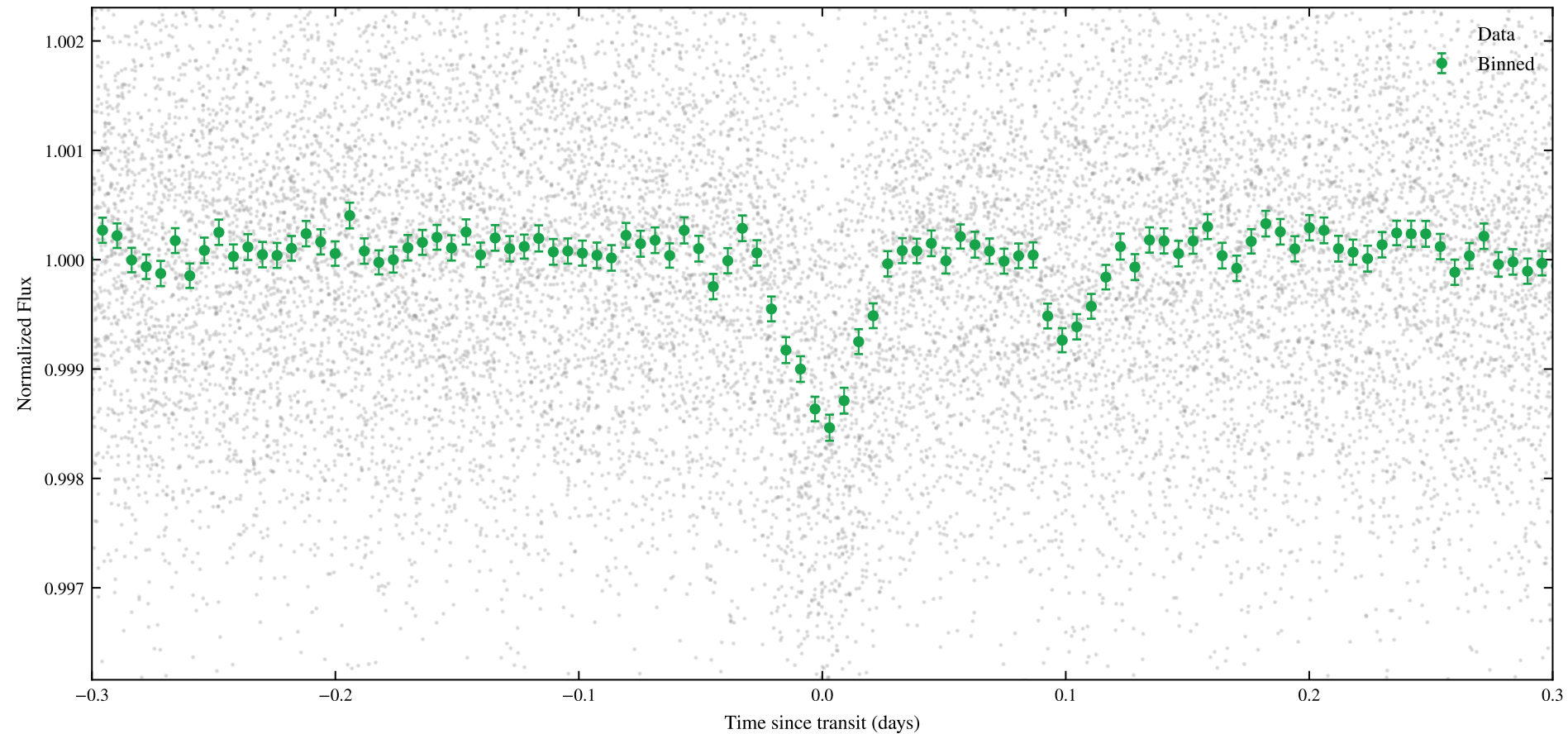
TIC 136274063 | Sectors: 102, 103 | MCMC Trace Plots



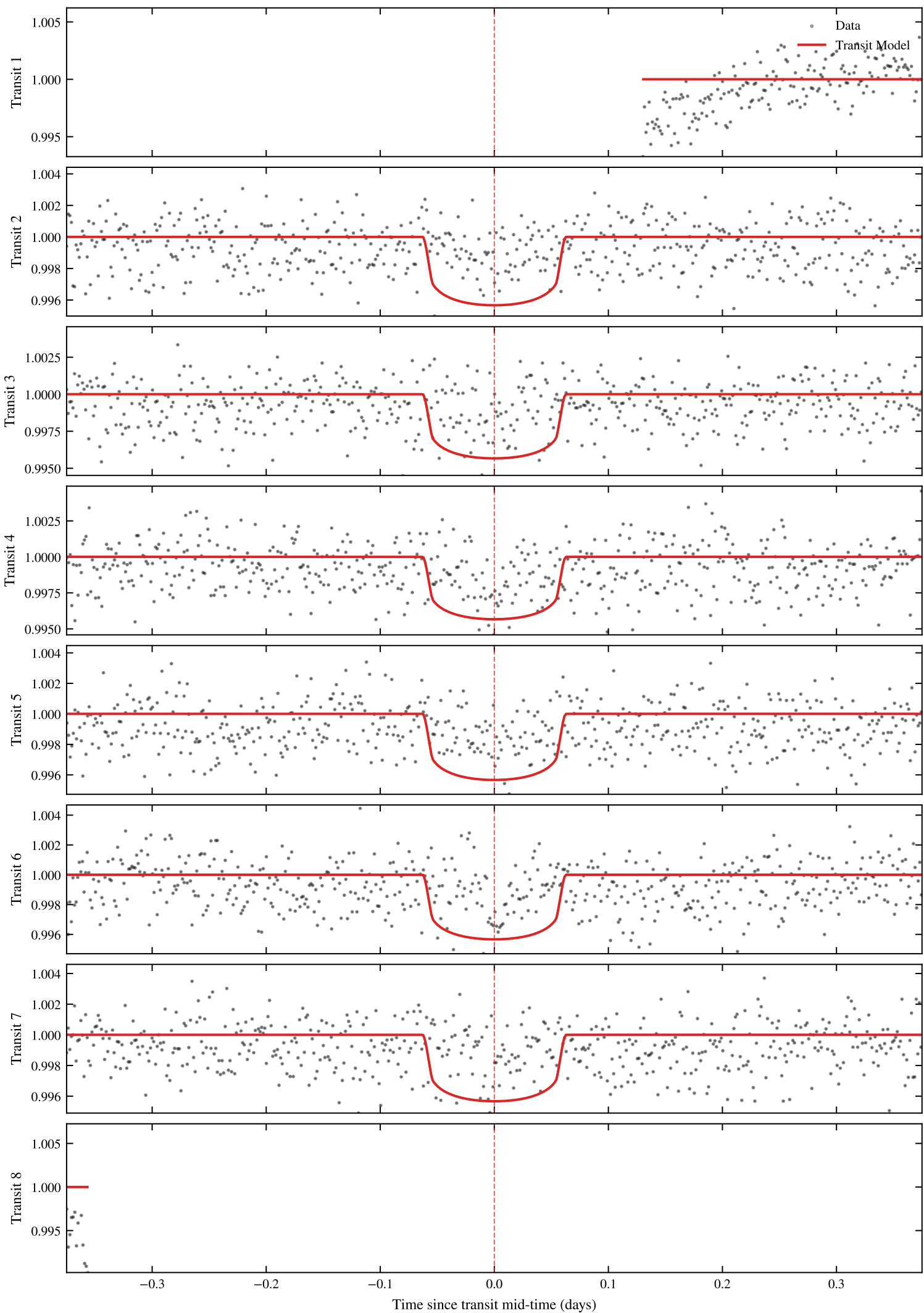
TIC 136274063 | Sectors: 102, 103 | Phase-Folded Transit (P = 0.59713 d)



TIC 136274063 | Sectors: 102, 103 | Phase-Folded Transit (P = 1.19452 d)



TIC 136274063 | Sectors: 102, 103 | Stacked Individual Transits



TIC 136274063 | Sectors: 102, 103 | Phased MCMC Fit - Planet 1

